SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY
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Module 02: Statistical Inference 1

Lecture 05: Point Estimation: Asymptotics

Asymptotics

Why Study Asymptotics and Consistency?

- In the theory of estimation, we want estimators that get **closer to the true parameter** as sample size increases.
- As n grows, estimates become **more tightly clustered around the population parameter** θ. This stability is essential in inference.
- A **consistent estimator** converges in probability to the true value as $n \to \infty$.

Example

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random sample from a population X with mean μ and variance σ^2.

$$\mathbb{E}[X_i] = \mu, \quad \mathsf{Var}(X_i) = \sigma^2. \text{ for all } i = 1, 2, \ldots, n.$$

Define the sample mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Then,

$$\mathsf{Var}(\bar{X}) = \mathrm{Var}\left(\frac{\sum_{i=1}^{n} X_i}{n}\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{\sigma^2}{n}$$

Since $\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = n\mathrm{Var}(X_i) = n\sigma^2$

Therefore, The sample mean $\overline{X}$ estimates μ with variance $\mathrm{Var}(\overline{X}) = \sigma^2/n \to 0$ as $n \to \infty$.

Consistency

Consistent Estimator

A sequence $\{T_n\}_{n \geq 1}$ of estimators (or simply an estimator T_n) is said to be a consistent estimator (weakly-consistent estimator) of θ if T_n converges to θ in probability, which means for every $\epsilon > 0$, we have $\lim_{n \to \infty} P[\,|T_n - \theta| < \epsilon] \to 1$

Note:
This is equivalent to: For every $\epsilon > 0$ and $\eta > 0$, there exists $n \geq m$ such that $P[\,|T_n - \theta| < \epsilon] > 1 - \eta$ whenever $n \geq m$, where m is a large value of n.

Strongly Consistent Estimator

An estimator T_n of a parameter θ is said to be strongly consistent if it converges to θ almost surely.
Which means that for every $\epsilon > 0$, we have $\lim_{n \to \infty} P[\,|T_n - \theta| < \epsilon] = 1$.

Sufficiency Condition for Consistency

If $\{T_n\}_{n\geq 1}$ is a sequence of estimators (or simply an estimator T_n) is such that for all $\theta \in \Theta$ then the estimator T_n is a consistent estimator if :

1. The estimator T_n is an asymptotically unbiased or simply unbiased estimator of θ. Formally, $E(T_n) \to \theta$ as $n \to \infty$.
2. The variance of estimator T_n decreases with increase in population size. In other words, $\mathrm{Var}(T_n) \to 0$ as $n \to \infty$.

Example 1

The sample mean is a consistent estimator of the population mean.

Proof: Let $X_1, X_2, \ldots, X_n$ be a random sample taken from a population X with mean $\mu < \infty$ and variance $\sigma^2 < \infty$.

For every $\epsilon > 0$, consider $\lim_{n\to\infty} P[|\overline{X} - \mu| < \epsilon] = \lim_{n\to\infty} P\left[\left|\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}\right| < \frac{\epsilon\sqrt{n}}{\sigma}\right]$

Example 1

Proof Continued: By Central Limit Theorem, $Z = \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ for large values of n.

Therefore, $\lim\limits_{n\to\infty} P[|\overline{X} - \mu| < \epsilon] = \lim\limits_{n\to\infty} P\left[|Z| < \dfrac{\epsilon\sqrt{n}}{\sigma}\right]$

$$= \lim_{n\to\infty} \int_{-\frac{\epsilon\sqrt{n}}{\sigma}}^{\frac{\epsilon\sqrt{n}}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1.$$

Example 2

The sample median is a consistent estimator of the population mean for a Normal Population.

Proof: Let X' be the sample median of a population $X \sim \mathcal{N}(\mu, \sigma^2)$.

For large values of n, it is known that:

$$E(X') = \mu \text{ and } \mathrm{Var}(X') = \frac{\pi\sigma}{2\sqrt{n}}.$$

Since X' is an unbiased estimator of μ and $\mathrm{Var}(X') \to 0$ as $n \to \infty$.

By the sufficient criteria for consistency, X' is a consistent estimator of μ. (Proved)

Sample Variance

Let $X_1, X_2, \ldots, X_n$ be a random sample from a population X with mean $\mu < \infty$ and $\sigma^2 < \infty$.
Both the expressions:
$S^2 = \dfrac{\sum_{i=1}^n (X_i - \overline{X})^2}{n-1}$ and $S'^2 = \dfrac{\sum_{i=1}^n (X_i - \overline{X})^2}{n}$ are called the sample variance depending upon the context.
However, the expression $S^2 = \dfrac{\sum_{i=1}^n (X_i - \overline{X})^2}{n-1}$ is is more widely accepted as the correct expression of sample variance.

Example 3

The estimator $S'^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}$ is a consistent estimator of the population variance.

Proof: Let $X_1, X_2, \ldots, X_n$ be a random sample taken from a population X with mean $\mu < \infty$ and variance $\sigma^2 < \infty$. Consider

$$E(S'^2) = E\left(\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}\right)$$

$$= \frac{1}{n}E\left(\sum_{i=1}^{n}(X_i - \overline{X})^2\right)$$

$$= \frac{1}{n}E((n-1)S^2)$$

where $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$.

Example 3

Proof Continued: Multiplying and dividing by σ^2,

we get $E(S'^2) = \dfrac{\sigma^2}{n} E\left(\dfrac{(n-1)S^2}{\sigma^2}\right)$.

However, $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ and hence

$$\begin{aligned} E(S'^2) &= \frac{\sigma^2}{n} E\left(\frac{(n-1)S^2}{\sigma^2}\right) \\ &= \frac{\sigma^2}{n} E(\chi^2_{n-1}) \\ &= \frac{\sigma^2(n-1)}{n}. \end{aligned}$$

Therefore, as $n \to \infty$ we have $E(S'^2) \to \sigma^2$.

Example 3

Proof Continued: Consider

$$\begin{aligned}
\mathrm{Var}(S'^2) &= \mathrm{Var}\left(\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}\right) \\
&= \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n}(X_i - \overline{X})^2\right) \\
&= \frac{1}{n^2}\mathrm{Var}((n-1)S^2).
\end{aligned}$$

Multiplying and dividing by σ^4, we get $\mathrm{Var}(S'^2) = \dfrac{\sigma^4}{n^2}\mathrm{Var}\left(\dfrac{(n-1)S^2}{\sigma^2}\right)$.

Example 3

Proof Continued: $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ and hence

$$\begin{aligned}
\mathrm{Var}(S'^2) &= \frac{\sigma^4}{n^2}\mathrm{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) \\
&= \frac{\sigma^4}{n^2}\mathrm{Var}(\chi^2_{n-1}) \\
&= \frac{2(n-1)\sigma^4}{n}.
\end{aligned}$$

Therefore, as $n \to \infty$ we have $\mathrm{Var}(S'^2) \to 0$.
By the sufficient criteria for consistency, S'^2 is a consistent estimator of σ^2. (Proved)

Example 4

An unbiased estimator which is not consistent.

Let $X_1, X_2, \ldots, X_n$ be a random sample taken from a population $X \sim \mathcal{N}(\mu, \sigma^2)$.
Consider the estimator $T = X_1$ for the population mean μ.
$E(X_1) = \mu$ and $\mathrm{Var}(X_1) = \sigma^2$.
Therefore, X_1 is an unbiased estimator of μ.

Example 4

However, we have:

$$P(|T - \mu| \geq \epsilon) = P(|X_1 - \mu| \geq \epsilon)$$

Since $X_1 \sim N(\mu, \sigma^2)$, this probability does not depend on n and is positive for any $\epsilon > 0$.

$$\text{In other words: } P(|X_1 - \mu| \geq \epsilon) = 1 - P(|X_1 - \mu| < \epsilon) > 0$$

Therefore, $T = X_1$ is not a consistent estimator of μ. (Proved)

Properties of Consistent Estimators

The following are important properties of consistent estimators:

- Consistent estimators may not be unique (Both sample mean and sample median were shown to be consistent estimators of population mean in Example 1 and Example 2).
- Consistent estimators need not be unbiased (The estimator S'^2 in Example 3 was consistent but not unbiased).
- An unbiased estimator need not be consistent (The estimator T in Example 4 was shown to be unbiased but not consistent).

Properties of Consistent Estimators

- **Invariance Property:** If T_n is an estimator of θ and $f(\theta)$ is a continuous function of θ. Then, $f(T_n)$ is a consistent estimator of $f(\theta)$.
- If T_n and S_n are consistent estimators of parameters α and β respectively. Then, the following hold:
    1. $T_n + S_n$ is a consistent estimator of $\alpha + \beta$.
    2. $T_n S_n$ is a consistent estimator of $\alpha\beta$.
    3. If $\beta \neq 0$ then T_n / S_n is a consistent estimator of α/β.

Drawbacks of Consistent Estimators

The following are the main drawbacks of consistent estimators:

- The shape of the sampling distribution of consistent estimators is not known as the estimators in most cases are non linear functions of the random samples.
- Even if the sampling distribution of the consistent estimator is known then its shape changes with sample size.

Consistent Asymptotically Normal Estimator

An estimator T_n is said to be a consistent asymptotically normal estimator for the parameter θ if the sampling distribution of $\sqrt{n}(T_n - \theta)$ follows $\mathcal{N}(0, \sigma^2)$.

References

1. https://www.egyankosh.ac.in/bitstream/123456789/104777/3/Unit-7.pdf
2. Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.


Thank you

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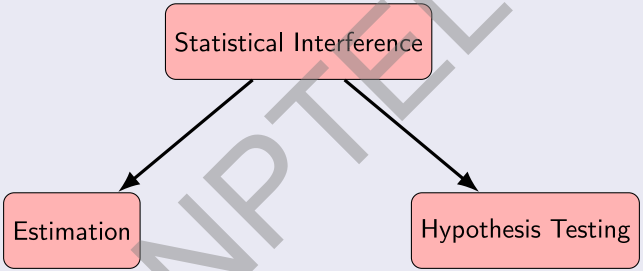
Module 02: Statistical Inference 1

Lecture 01: Point Estimation: Unbiased Estimation

Statistical Inference



Estimation

Tries to predict a value related to the population using a sample statistic $T(X_1, \ldots, X_n)$.

- Population: Follows the distribution of some random variable X.
- Sample: $X_1, \ldots, X_n$ such that $F_{X_i} = F_X$ for all i.
- Sample Statistic: A function of sample $T(X_1, \ldots, X_n)$.

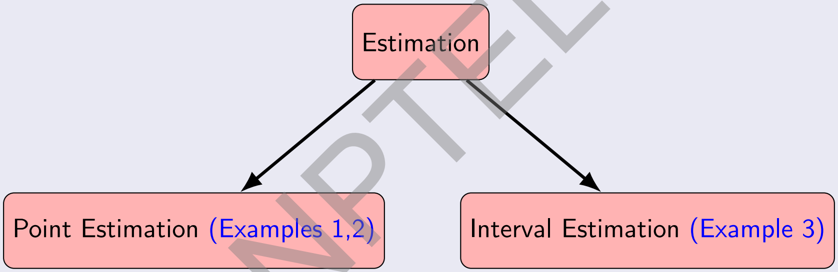
Estimation (continued)

Example:

1. Predict the head probability of a coin using the number of heads in n tosses.
2. Predict the mean of a Normal Distribution population by $\frac{X_1+\cdots+X_n}{n}$ where $X_1, \ldots, X_n$ is a sample from that.
3. Try to predict an interval within the variance of a normal distribution that is very likely to belong using $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$

   where $X_1, \ldots, X_n$ is the sample and $\bar{X} = \sum_{i=1}^{n} X_i$, the sample mean.

Estimation (continued)



Hypothesis testing

Tries to decide between two given options about the population which one is more likely.

Example:

- Based on the results of n tosses, try to decide for two given values P_1 and P_2, which is more likely to be the head probability.
- Based on a sample $X_1, \ldots, X_n$ of the normal population $\mathcal{N}(0, \sigma^2)$, decide between two given values σ_1^2 and σ_2^2.

Independence

- Two sets A and B are called independent if $P(A \cap B) = P(A)P(B)$.
- Random variables $X_1, \ldots, X_n$ are called independent if $P(X_1 \in A_1, \ldots X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i)$.
- Unless otherwise stated for a sample $X_1, \ldots, X_n$ are independent.
- So for a sample $X_1, \ldots, X_n$ foe population with cdf F_X (for some random variable X), we have $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) with cdf F_X.

Point Estimation

1. Unbiasedness.
2. Likelihood.
3. Method of Moments.
4. Asymptotic behaviour.

Unbiasedness

Let θ be a value related to F_x and $T(X_1, \ldots, X_n)$ is a function of sample.
$T(X_1, \ldots, X_n)$ is called an *estimator* or a *predictor* of θ if it is an "educated guess" for θ.
A predictor $T(X_1, \ldots, X_n)$ of a value θ associated to a population using sample $X_1, \ldots, X_n$ is called *Unbiased* if $E(T) = \theta$.

Facts

- For $E(\sum_{i=1}^{n} Z_i) = \sum_{i=1}^{n} E(Z_i)$
- $E(cZ) = cE(Z)$ for any constant c.
- For pairwise independent random variable $Z_1, \ldots, Z_n$,

$$\mathrm{Var}(\sum_{i=1}^{n} c_i Z_i) = \sum_{i=1}^{n} {c_i}^2 \mathrm{Var}(Z_i) \text{ where } c_1, \ldots, c_n \text{ are scalars.}$$

Example

Sample Mean for Finite Population :

- $X_1, \ldots, X_N$ finite population and
- $Y_1, \ldots, Y_n$ is i.i.d. sample. Then $E(\bar{Y}) = \bar{X}$.

Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$.

So

$$
\begin{aligned}
E(\bar{Y}) &= E\left(\frac{\sum_{i=1}^{n} Y_i}{n}\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} E(Y_i) \\
&= E(Y_1) && \text{(because } \mathrm{Y_i's} \text{ are i.i.d.)} \\
&= \frac{1}{N}\sum_{j=1}^{N} X_j \\
&= \bar{X} && \text{(Proved).}
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. Ber(p), then $E(\bar{X}) = p$.

  Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

  As $X_1, \ldots, X_n$ follow $Ber(p)$ distribution, therefore, $E(X_1) = \cdots = E(X_n) = p$.
  So

$$
\begin{aligned}
E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\
&= \frac{1}{n}(p + \cdots + p) \\
&= p
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$, then $E(\bar{X}) = \mu$.

Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

As $X_1, \ldots, X_n$ follow $\mathcal{N}(\mu, \sigma^2)$ distribution , therefore, $E(X_1) = \cdots = E(X_n) = \mu$
So

$$
\begin{aligned}
E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\
&= \frac{1}{n}(\mu + \cdots + \mu) \\
&= \mu
\end{aligned}
$$

Examples

Sample Variance for Finite Population :

- $X_1, \ldots, X_N$ finite population and
- $Y_1, \ldots, Y_n$ is i.i.d. sample. Find the unbiased estimator for $\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2$.

Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$. Let $s^2 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$.

Now,

$$
\begin{aligned}
\sum_{i=1}^{n}(Y_i - \bar{X})^2 &= \sum_{i=1}^{n}((Y_i - \bar{Y}) + (\bar{Y} - \bar{X}))^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2\sum_{i=1}^{n}(Y_i - \bar{Y})(\bar{Y} - \bar{X}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})\sum_{i=1}^{n}(Y_i - \bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})(n\bar{Y} - n\bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + n(\bar{Y} - \bar{X})^2
\end{aligned}
$$

Examples (continued)

Then,

$$
\begin{aligned}
E(s^2) &= E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2 - (\bar{Y} - \bar{X})^2\right]
\end{aligned}
$$

Now

$$
\begin{aligned}
& E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}\frac{1}{n}\sum_{i=1}^{n}Y_i + \bar{X}^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}^2 + \bar{X}^2\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}E(Y_i^2) - \bar{X}^2 \\
&= E(Y_1^2) - \bar{X}^2
\end{aligned}
$$

Examples (continued)

Also

$$\begin{aligned} & E(\bar{Y} - \bar{X})^2 \\ & = E(\bar{X}^2 + \bar{Y}^2 - 2\bar{X}\bar{Y}) \\ & = E(\bar{Y}^2) - \bar{X}^2 \end{aligned}$$

So

$$E(s^2) = E({Y_1}^2) - E(\bar{Y}^2)$$

Examples (continued)

$$
\begin{aligned}
E(\bar{Y}^2) &= E\left(\frac{Y_1 + \cdots + Y_n}{n}\right)^2 \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E({Y_i}^2) + 2E\left(\sum_{1\le i_1 < j_1 \le n} Y_{i_1} Y_{j_1}\right)\right\} \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E({Y_i}^2) + 2\sum_{1\le i_1 < j_1 \le n} E(Y_{i_1})E(Y_{j_1})\right\} \\
&= \frac{1}{n^2}\left\{\frac{n}{N}\sum_{j=1}^{N} {X_j}^2 + \frac{2n(n-1)}{2}\bar{X}^2\right\} \\
&= \frac{1}{nN}\sum_{j=1}^{N} {X_j}^2 + \frac{(n-1)}{n}\bar{X}^2
\end{aligned}
$$

Examples (continued)

Remember

$$E(Y_1{}^2) = \frac{1}{N}\sum_{j=1}^{N} X_j{}^2$$

So,

$$\begin{aligned} E(s^2) &= \frac{1}{N}\sum_{j=1}^{N} X_j{}^2 - \frac{1}{nN}\sum_{j=1}^{N} X_j{}^2 - \frac{(n-1)}{n}\bar{X}^2 \\ &= \frac{n-1}{n}\frac{1}{N}\sum_{j=1}^{N} X_j{}^2 - \frac{(n-1)}{n}\bar{X}^2 \\ &= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N} X_j{}^2 - \bar{X}^2\right\} \\ &= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2\right\} \end{aligned}$$

Examples (continued)

Observe that

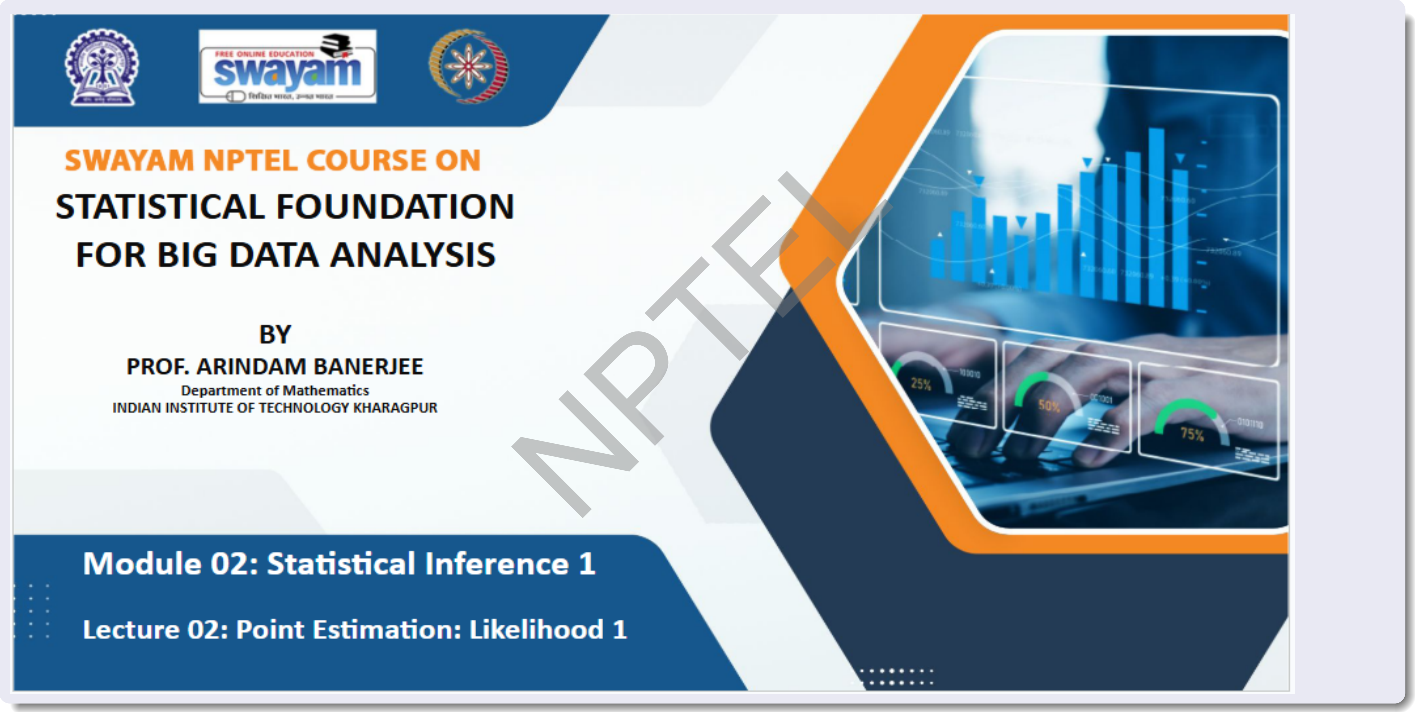
$$\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j-\bar{X})^2\right\} = \frac{n}{n-1}E(s^2)$$
$$= E\left(\frac{n}{n-1}s^2\right)$$
$$= E\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i-\bar{Y})^2\right]$$

So the unbiased estimator of $\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j-\bar{X})^2\right\}$ is $\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i-\bar{Y})^2\right]$

References

1. *Fundamentals of Mathematical Statistics by* by S.C. Gupta and V.K. Kapoor, 10th revised edition, Sultan Chand and Sons
2. AN INTRODUCTION TO PROBABILITY AND STATISTICS by VIJAY K. ROHATGI A. K. Md. EHSANES SALEH, Third Edition, WILEY.
3. A FIRST COURSE IN PROBABILITY by Sheldon Ross, Eighth Edition, Pearson


Thank you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Lecture 02: Point Estimation: Likelihood 1
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Example

Sample Mean for Finite Population :

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Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$.
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\begin{aligned}
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&= \frac{1}{n}\sum_{i=1}^{n} E(Y_i) \\
&= E(Y_1) && \text{(because } Y_i'\text{s are i.i.d.)} \\
&= \frac{1}{N}\sum_{j=1}^{N} X_j \\
&= \bar{X} && \text{(Proved).}
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. Ber(p), then $E(\bar{X}) = p$.

  Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
  As $X_1, \ldots, X_n$ follow $Ber(p)$ distribution, therefore, $E(X_1) = \cdots = E(X_n) = p$.
  So

$$
\begin{aligned}
E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\
&= \frac{1}{n}(p + \cdots + p) \\
&= p
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$, then $E(\bar{X}) = \mu$.

  Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
  As $X_1, \ldots, X_n$ follow $\mathcal{N}(\mu, \sigma^2)$ distribution , therefore, $E(X_1) = \cdots = E(X_n) = \mu$
  So

$$\begin{aligned} E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\ &= \frac{1}{n}(\mu + \cdots + \mu) \\ &= \mu \end{aligned}$$

Examples

Sample Variance for Finite Population :

- $X_1, \ldots, X_N$ finite population and
- $Y_1, \ldots, Y_n$ is i.i.d. sample. Find the unbiased estimator for $\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2$.

Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$. Let $s^2 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$.
Now,

$$\begin{aligned}
\sum_{i=1}^{n}(Y_i - \bar{X})^2 &= \sum_{i=1}^{n}((Y_i - \bar{Y}) + (\bar{Y} - \bar{X}))^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2\sum_{i=1}^{n}(Y_i - \bar{Y})(\bar{Y} - \bar{X}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})\sum_{i=1}^{n}(Y_i - \bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})(n\bar{Y} - n\bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + n(\bar{Y} - \bar{X})^2
\end{aligned}$$

Examples (continued)

Then,

$$
\begin{aligned}
E(s^2) &= E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2 - (\bar{Y} - \bar{X})^2\right]
\end{aligned}
$$

Now

$$
\begin{aligned}
& E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}\frac{1}{n}\sum_{i=1}^{n}Y_i + \bar{X}^2\right] \\
&= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}^2 + \bar{X}^2\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}E(Y_i^2) - \bar{X}^2 \\
&= E(Y_1^2) - \bar{X}^2
\end{aligned}
$$

Examples (continued)

Also

$$
\begin{aligned}
& E(\bar{Y}-\bar{X})^2 \\
& = E(\bar{X}^2+\bar{Y}^2-2\bar{X}\bar{Y}) \\
& = E(\bar{Y}^2)-E(\bar{X}^2)
\end{aligned}
$$

So

$$E(s^2) = E({Y_1}^2) - E(\bar{Y}^2)$$

Examples (continued)

$$
\begin{aligned}
E(\bar{Y}^2) &= E\left(\frac{Y_1 + \cdots + Y_n}{n}\right)^2 \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E(Y_i{}^2) + 2E\left(\sum_{1\le i_1<j_1\le n} Y_{i_1}Y_{j_1}\right)\right\} \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E(Y_i{}^2) + 2\sum_{1\le i_1<j_1\le n} E(Y_{i_1})E(Y_{j_1})\right\} \\
&= \frac{1}{n^2}\left\{\frac{n}{N}\sum_{j=1}^{N} X_j{}^2 + \frac{2n(n-1)}{2}\bar{X}^2\right\} \\
&= \frac{1}{nN}\sum_{j=1}^{N} X_j{}^2 + \frac{(n-1)}{n}\bar{X}^2
\end{aligned}
$$

Examples (continued)

Remember

$$E({Y_1}^2) = \frac{1}{N}\sum_{j=1}^{N} {X_j}^2$$

So,

$$\begin{aligned}
E(s^2) &= \frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \frac{1}{nN}\sum_{j=1}^{N} {X_j}^2 - \frac{(n-1)}{n}\bar{X}^2 \\
&= \frac{n-1}{n}\frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \frac{(n-1)}{n}\bar{X}^2 \\
&= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \bar{X}^2\right\} \\
&= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N} (X_j - \bar{X})^2\right\}
\end{aligned}$$

Examples (continued)

Observe that

$$\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j-\bar{X})^2\right\} = \frac{n}{n-1}E(s^2)$$
$$= E\left(\frac{n}{n-1}s^2\right)$$
$$= E\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i-\bar{Y})^2\right]$$

So the unbiased estimator of $\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j-\bar{X})^2\right\}$ is $\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i-\bar{Y})^2\right]$

Motivation

- Suppose we toss a coin with head probability p and we get a head.
- Between the following two options, which is more LIKELY ?
  1. p=0.7,
  2. p=0.9

Example

Number of toss	Number of head	option 1 for p	option 2 for p	Which one is more likely
100	90	0.2	0.8	option 2
10	1	0.2	0.5	option 1
100	100	0.6	0.9	option 2
20	4	0.8	0.3	option 2

Maximal Likelihood Estimator: Definitions

- Let $X_1, \ldots, X_n$ be i.i.d. with pdf $f_\theta(x)$.
- The *likelihood function* is defined by

$$L = L(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

- The *maximum likelihood estimator* or MLE, denoted by $\hat{\theta}$ is the value of θ that maximizes L.

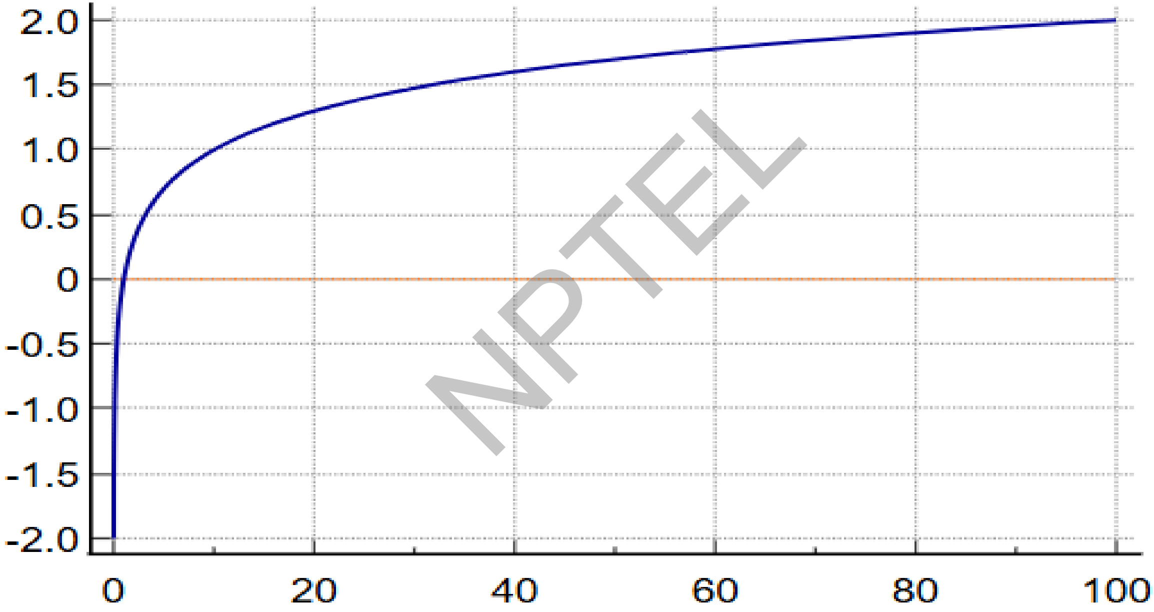
Maximal Likelihood Estimator: Definitions (continued)

- Let $X_1, \ldots, X_n$ be i.i.d. with pmf $P_\theta(X)$.
- The *likelihood function* is defined by

$$L = L(\theta) = \prod_{i=1}^{n} P_\theta(X_i).$$

- The *maximum likelihood estimator* or MLE, denoted by $\hat{\theta}$ is the value of θ that maximizes L.

Graph of the logarithmic function



Maximum Likelihood Estimation for the Exponential Distribution

- Recall that the pdf of the exponential distribution is given by

$$f_\theta(x) = \frac{1}{\theta} e^{\frac{-x}{\theta}}$$

  where $x \in [0, \infty)$ and $\theta > 0$
- Let $X_1, \ldots, X_n$ be random samples from exponential distribution.
- Our target is to estimate the parameter θ.

Maximum Likelihood Estimation for the Exponential Distribution (continued)

- The likelihood function is given by

$$
\begin{aligned}
L(\theta) &= \prod_{i=1}^{n} f(X_i;\theta) \\
&= \frac{1}{\theta^n} e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i}
\end{aligned}
$$

- Taking log both sides we have

$$
\begin{aligned}
\log(L(\theta)) &= \log(\theta^{-n}) + \log(e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i}) \\
&= -n\log(\theta) - \frac{1}{\theta}\sum_{i=1}^{n} X_i
\end{aligned}
$$

Maximum Likelihood Estimation for the Exponential Distribution (continued)

- Now taking the partial derivative with respect to θ and set that equal to 0, we have

$$-\frac{n}{\theta} + \frac{\sum_{i=1}^{n} X_i}{\theta^2} = 0$$

$$\text{or, } \theta = \frac{\sum_{i=1}^{n} X_i}{n} = \bar{X}$$

- Therefore the maximum likelihood estimate of θ is $\hat{\theta} = \bar{X}$ and this gets verified by the second derivative test.

Maximum Likelihood Estimation for the Exponential Distribution (continued)

Also note that

$$\begin{aligned}\frac{\partial^2 \log(L)}{\partial\theta^2} &= \frac{n}{\theta^2} - \frac{2\sum_{i=1}^{n} X_i}{\theta^3}\\ &= \frac{n\theta - 2\sum_{i=1}^{n} X_i}{\theta^3}\end{aligned}$$

Now when $\theta = \hat{\theta} = \bar{X}$, then the above expression yields,

$$\frac{\partial^2 \log(L(\theta))}{\partial\theta^2} = \frac{-n\bar{X}}{\theta^3} < 0$$

Therefore θ attains maximum at $\hat{\theta} = \bar{X}$

Examples: Continuous Distribution

Likelihood Estimation for Normal Distribution

- The pdf of the $\mathcal{N}(\mu, \sigma^2)$ distribution is given by

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

for all $x \in \mathbb{R}$.

- Let $X_1, \ldots, X_n$ be a sample from $\mathcal{N}(\mu, \sigma^2)$, where both μ and σ^2 are unknown and $\theta = (\mu, \sigma^2)$.
- The likelihood function is

$$\begin{aligned} L(\mu, \sigma^2) \\ = \prod_{i=1}^{n} f_\theta(X_i) \\ = \frac{1}{\sigma^n (2\pi)^{n/2}} e^{\left\{-\sum_{i=1}^{n} \frac{(X_i-\mu)^2}{2\sigma^2}\right\}} \end{aligned}$$

Likelihood Estimation for Normal Distribution (continued)

- Taking log both sides we obtain

$$\log(L(\mu,\sigma^2)) = -\frac{n}{2}\log\sigma^2 - \frac{n}{2}\log(2\pi) - \left\{\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{2\sigma^2}\right\}$$
$$= -n\log(\sigma) - \frac{n}{2}\log(2\pi) - \left\{\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{2\sigma^2}\right\}.$$

- The likelihood equations are obtained by setting $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\mu} = 0$ and $\frac{\partial(\log L)}{\partial\sigma^2} = 0$

Likelihood Estimation for Normal Distribution (continued)

- $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\mu} = 0$ gives

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i-\mu)=0$$
$$\text{or, } (X_1+\cdots+X_n)-n\mu=0$$
$$\text{or, } \mu=\frac{X_1+\cdots+X_n}{n}$$

We get $\hat{\mu}=\bar{X}$

Likelihood Estimation for Normal Distribution (continued)

- To estimate σ^2, we need to set $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\sigma^2} = 0$ i,e,

$$\frac{-n}{2}\frac{1}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i-\mu)^2 = 0$$

$$\text{or, } \frac{n}{2} = \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2$$

$$\text{or, } \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2$$

- Thus the likelihood estimations of μ and σ^2 are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2$ respectively.
- One can check through the second derivative test that μ and σ^2 attain maximum at $\hat{\mu}$ and $\hat{\sigma}^2$

Maximum Likelihood Estimation for the Uniform $(0, \theta)$ Distribution

- Recall that the PDF is

$$f_\theta(x) = \begin{cases} \frac{1}{\theta}, & \text{if } 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

- Let $X_1, \ldots, X_n \sim \mathrm{Unif}(0, \theta)$.
- Consider a fixed value of θ. Suppose $\theta < X_i$ for some i. Then $f_\theta(x_i) = 0$ and hence the Likelihood function

$$L = \prod_{i=1}^{n} f_\theta(X_i) = 0.$$

Maximum Likelihood Estimation for the Uniform $(0, \theta)$ Distribution (continued)

- It follows that $L = 0$ if any $X_i > \theta$. Therefore $L = 0$ if $\theta < X_{(n)}$, where $X_{(n)} = \max\{X_1, \ldots, X_n\}$.
- Now consider $X_{(n)} \leq \theta$. For every X_i we have that $f_\theta(X_i) = 1/\theta$.
- The likelihood estimator

$$L(\theta(x)) = \begin{cases} \frac{1}{\theta^n}, & \text{if } 0 \leq X_{(n)} \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

- Hence $\hat{\theta} = X_{(n)}$ maximizes it, so it is MLE.

Maximum Likelihood Estimation for the Gamma Distribution

- Recall that the pdf of the Gamma distribution can be written as

$$f_\theta(x) = \frac{1}{\Gamma(p)\theta^p} x^{p-1} e^{\frac{-x}{\theta}}$$

where $x \in [0, \infty)$ and $p, \theta > 0$.
- Let $X_1, \dots, X_n$ be random sample for the Gamma distribution.
- Our task is to derive the MLE of the parameter θ if p is known. where $x, p, \theta > 0$.

Maximum Likelihood Estimation for the Gamma Distribution

- In this case the likelihood function is

$$
\begin{aligned}
L(\theta) &= \prod_{i=1}^{n} f_\theta(X_i) \\
&= \prod_{i=1}^{n} \frac{1}{\Gamma(p)\theta^p} {X_i}^{p-1} e^{\frac{-X_i}{\theta}} \\
&= \theta^{-np}\Gamma(p)^{-n} e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i} \prod_{i=1}^{n} {X_i}^{p-1}
\end{aligned}
$$

Maximum Likelihood Estimation for the Gamma Distribution (continued)

- Taking log both sides we have,

$$\log(L(\theta)) = -np\log(\theta) - n\log(\Gamma(p)) - \frac{1}{\theta}\sum_{i=1}^{n} X_i + \log\prod_{i=1}^{n} {X_i}^{p-1}$$

- Now setting $\frac{\partial \log(L(\theta))}{\partial\theta} = 0$ we get

$$\frac{-np}{\theta} + \frac{\sum_{i=1}^{n} X_i}{\theta^2} = 0$$
$$\text{or, } \frac{\sum_{i=1}^{n} X_i}{np} = \theta$$

  i.e, $\hat{\theta} = \frac{\bar{X}}{p}$ is the MLE of θ.

- One can check through second derivative test that θ attains maximum at $\hat{\theta}$.

Likelihood Estimation for Discrete Distribution

Likelihood Estimation for the Poisson μ population

- The pdf of the Poisson μ population

$$P_\mu(X = x) = e^{-\mu}\frac{\mu^x}{x!}$$

where $x \in \mathbb{N}_0$.

- Let $X_1, \ldots, X_n$ be sample from Poisson μ population.
- The likelihood function is

$$\begin{aligned} L(\mu) &= \prod_{i=1}^{n} P_\mu(X_i) \\ &= e^{-n\mu}\frac{\mu^{n\bar{X}}}{X_1! \ldots X_n!} \end{aligned}$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Likelihood Estimation for the Poisson μ population (continued)

- Taking log both sides,

$$\log(L(\mu)) = -n\mu + n\bar{X}\log(\mu) - \log(X_1! \dots X_n!).$$

- Now taking the partial derivative with respect to μ and setting that equal to zero we get

$$-n + \frac{n\bar{X}}{\mu} = 0$$
$$\text{or, } \bar{X} = \mu$$

- Thus the maximum likelihood estimate for μ is $\hat{\mu} = \bar{X}$.
- One can check through second derivative test that μ attains maximum at $\hat{\mu}$.

Likelihood Estimation for the Bernoulli Distribution

- Recall that the probability mass function of the Bernoulli (p) is

$$P_p(X = x) = p^x(1-p)^{1-x}$$

for $x = 0, 1$ where p is the probability of success in each trial and the unknown parameter is p.

- Let $X_1, \ldots, X_n \sim$ Bernoulli (p) where p is the probability of success in each trial. The likelihood function is

$$L = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^S(1-p)^{n-S}$$

where $S = \sum_{i=1}^{n} X_i$.

Likelihood Estimation for the Bernoulli Distribution (continued)

- After taking log both sides,

$$\log(L) = S\log(p) + (n - S)\log(1 - p)$$

- Now take the derivative and set it equal to 0, we have

$$\begin{aligned} \frac{S}{p} - \frac{n-S}{1-p} &= 0 \\ \text{or, } (1-p)S - (n-S)p &= 0 \\ \text{or, } S - pS - np + pS &= 0 \\ \text{or, } S &= np \end{aligned}$$

- Therefore, $\hat{p} = \frac{S}{n} = \bar{X}$ which is the required MLE of p.
- One can check via second derivative test that p attains maximum at $\hat{p}$.

Likelihood Estimation for the Binomial (N, p) Distribution

- Recall that the pdf of the binomial (n, p) population is

$$P_p(X = x) = \binom{N}{x} p^x (1-p)^{N-x}$$

  N is the number of trials and p is the probability of success in each trial and x is the number of total successes.
- Let $X_1, \ldots, X_n \sim \mathrm{Bin}(N, p)$.
- Here our task is to estimate the parameter p.

Likelihood Estimation for the Binomial (N, p) Distribution (continued)

- Therefore the likelihood function is

$$L(p) = \prod_{i=1}^{n} P_p(x_i)$$
$$= \binom{N}{X_1} \cdots \binom{N}{X_n} p^{X_1+\cdots+X_n}(1-p)^{nN-(X_1+\cdots+X_n)}$$

- Taking log both sides

$$\log(L(p)) = (X_1 + \cdots + X_n)\log(p) + \{nN - (X_1 + \cdots + X_n)\}\log(1-p) + \log\left(\binom{N}{X_1} \cdots \binom{N}{X_n}\right)$$

Likelihood Estimation for the Binomial (N, p) Distribution (continued)

- Now $\frac{\partial \log(L(p))}{\partial p} = 0$ gives

$$
\begin{aligned}
\frac{X_1 + \cdots + X_n}{p} &= \frac{nN - (X_1 + \cdots + X_n)}{1 - p} \\
&= \frac{nN - (X_1 + \cdots + X_n) + (X_1 + \cdots + X_n)}{p + 1 - p} \\
&= nN \\
\text{or, } p &= \frac{(X_1 + \cdots + X_n)}{nN} \\
&= \frac{\bar{X}}{N}
\end{aligned}
$$

i,e, $\hat{p} = \frac{\bar{X}}{N}$ is the required maximum likelihood estimate of p.

- One can check through second derivative test that p attains maximum at $\hat{p}$.

Example

Question: Let us choose a sample X_1 of unit size from a population having pdf:

$$f_\alpha(x) = \begin{cases} \frac{2}{\alpha^2}(\alpha - x) & \text{if } 0 < x < \alpha \\ 0 & \text{otherwise} \end{cases}$$

Prove that the maximum likelihood estimate of the parameter α is $2X_1$.

Example (continued)

Solution: For a sample of unit size, the likelihood function is

$$L(\alpha) = f_\alpha(X_1) = \frac{2}{\alpha^2}(\alpha - X_1)$$

Taking log both sides of the above equation we get

$$\log(L(\alpha)) = \log(2) - 2\log(\alpha) + \log(\alpha - X_1)$$

Example (continued)

Now $\frac{\partial \log(L(\alpha))}{\partial \alpha} = 0$ gives

$$\begin{aligned} \frac{-2}{\alpha} + \frac{1}{\alpha - X_1} &= 0 \\ \text{or, } 2(\alpha - X_1) - \alpha &= 0 \\ \text{or, } \alpha &= 2X_1 \end{aligned}$$

Also

$$\frac{\partial^2 \log(L(\alpha))}{\partial^2 \alpha} = \frac{2}{\alpha^2} - \frac{1}{(\alpha - x)^2}$$

At $\alpha = X_1$, its numerator becomes

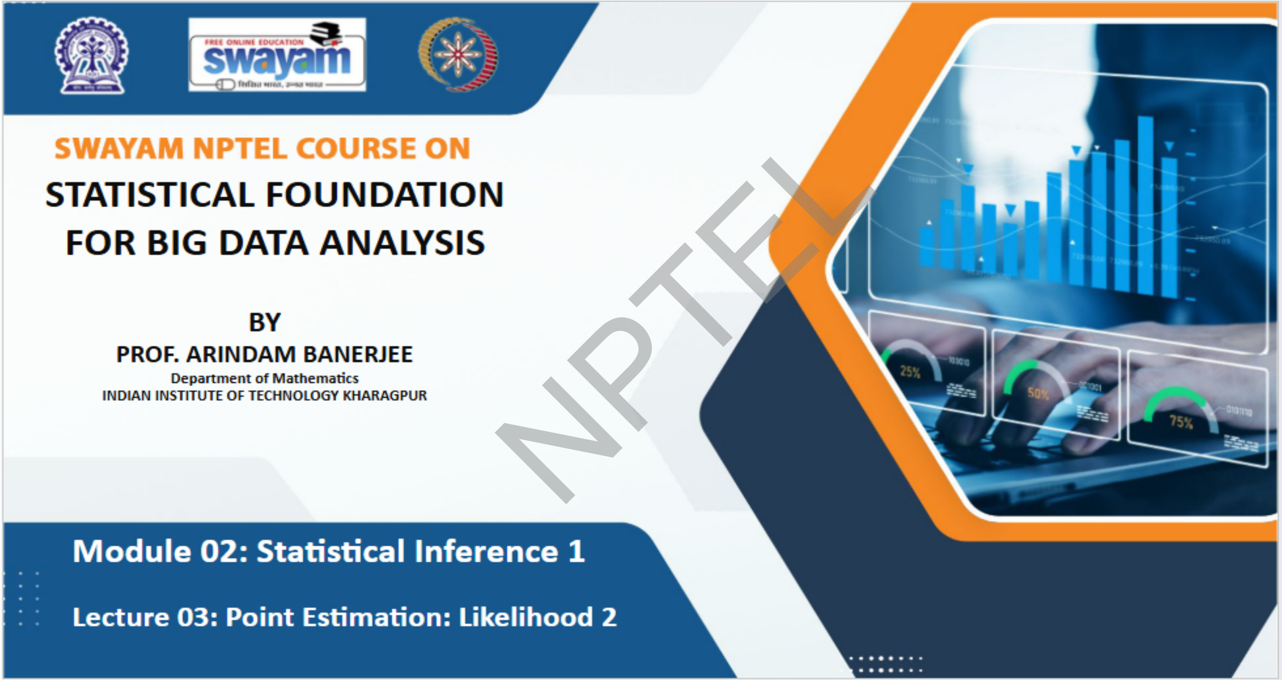
$$2(2X_1 - X_1)^2 - (2X_1)^2 = -2{X_1}^2 < 0$$

Hence the MLE of α is $\hat{\alpha} = 2X_1$.

References

1. Wikipedia
2. https://egrcc.github.io/docs/math/all-of-statistics.pdf
3. An Introduction To Probability And Statistics, 3rd edition by V.K. Rohatgi and A.K. Saleh, Wiley Series in Probability and Statistics.
4. Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.


Thank you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Example

Sample Mean for Finite Population :

- $X_1, \ldots, X_N$ finite population and
- $Y_1, \ldots, Y_n$ is i.i.d. sample. Then $E(\bar{Y}) = \bar{X}$.

Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$.
So

$$
\begin{aligned}
E(\bar{Y}) &= E\left(\frac{\sum_{i=1}^{n} Y_i}{n}\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} E(Y_i) \\
&= E(Y_1) && \text{(because } Y_i'\text{s are i.i.d.)} \\
&= \frac{1}{N}\sum_{j=1}^{N} X_j \\
&= \bar{X} && \text{(Proved).}
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. Ber(p), then $E(\bar{X}) = p$.

  Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
  As $X_1, \ldots, X_n$ follow $Ber(p)$ distribution, therefore, $E(X_1) = \cdots = E(X_n) = p$.
  So

$$
\begin{aligned}
E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\
&= \frac{1}{n}(p + \cdots + p) \\
&= p
\end{aligned}
$$

Example

- $X_1, \ldots, X_n \sim$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$, then $E(\bar{X}) = \mu$.

  Proof: We have $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
  As $X_1, \ldots, X_n$ follow $\mathcal{N}(\mu, \sigma^2)$ distribution , therefore, $E(X_1) = \cdots = E(X_n) = \mu$
  So

$$\begin{aligned} E(\bar{X}) &= \frac{1}{n}\sum_{i=1}^{n} E(X_i) \\ &= \frac{1}{n}(\mu + \cdots + \mu) \\ &= \mu \end{aligned}$$

Examples

Sample Variance for Finite Population :

- $X_1, \ldots, X_N$ finite population and
- $Y_1, \ldots, Y_n$ is i.i.d. sample. Find the unbiased estimator for $\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2$.

Proof: We have $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} X_j$. Let $s^2 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$.
Now,

$$
\begin{aligned}
\sum_{i=1}^{n}(Y_i - \bar{X})^2 &= \sum_{i=1}^{n}((Y_i - \bar{Y}) + (\bar{Y} - \bar{X}))^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2\sum_{i=1}^{n}(Y_i - \bar{Y})(\bar{Y} - \bar{X}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})\sum_{i=1}^{n}(Y_i - \bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + 2(\bar{Y} - \bar{X})(n\bar{Y} - n\bar{Y}) + n(\bar{Y} - \bar{X})^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + n(\bar{Y} - \bar{X})^2
\end{aligned}
$$

Examples (continued)

Then,

$$E(s^2) = E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right]$$
$$= E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2 - (\bar{Y} - \bar{X})^2\right]$$

Now

$$E\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{X})^2\right]$$
$$= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}\frac{1}{n}\sum_{i=1}^{n}Y_i + \bar{X}^2\right]$$
$$= E\left[\frac{1}{n}\sum_{i=1}^{n}Y_i^2 - 2\bar{X}^2 + \bar{X}^2\right]$$
$$= \frac{1}{n}\sum_{i=1}^{n}E(Y_i^2) - \bar{X}^2$$
$$= E(Y_1^2) - \bar{X}^2$$

Examples (continued)

Also

$$
\begin{aligned}
& E(\bar{Y}-\bar{X})^2 \\
& = E(\bar{X}^2+\bar{Y}^2-2\bar{X}\bar{Y}) \\
& = E(\bar{Y}^2)-E(\bar{X}^2)
\end{aligned}
$$

So

$$E(s^2) = E({Y_1}^2) - E(\bar{Y}^2)$$

Examples (continued)

$$
\begin{aligned}
E(\bar{Y}^2) &= E\left(\frac{Y_1 + \cdots + Y_n}{n}\right)^2 \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E(Y_i{}^2) + 2E\left(\sum_{1\le i_1<j_1\le n} Y_{i_1}Y_{j_1}\right)\right\} \\
&= \frac{1}{n^2}\left\{\sum_{i=1}^{n} E(Y_i{}^2) + 2\sum_{1\le i_1<j_1\le n} E(Y_{i_1})E(Y_{j_1})\right\} \\
&= \frac{1}{n^2}\left\{\frac{n}{N}\sum_{j=1}^{N} X_j{}^2 + \frac{2n(n-1)}{2}\bar{X}^2\right\} \\
&= \frac{1}{nN}\sum_{j=1}^{N} X_j{}^2 + \frac{(n-1)}{n}\bar{X}^2
\end{aligned}
$$

Examples (continued)

Remember

$$E({Y_1}^2) = \frac{1}{N}\sum_{j=1}^{N} {X_j}^2$$

So,

$$\begin{aligned} E(s^2) &= \frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \frac{1}{nN}\sum_{j=1}^{N} {X_j}^2 - \frac{(n-1)}{n}\bar{X}^2 \\ &= \frac{n-1}{n}\frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \frac{(n-1)}{n}\bar{X}^2 \\ &= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N} {X_j}^2 - \bar{X}^2\right\} \\ &= \frac{n-1}{n}\left\{\frac{1}{N}\sum_{j=1}^{N} (X_j - \bar{X})^2\right\} \end{aligned}$$

Examples (continued)

Observe that

$$\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2\right\} = \frac{n}{n-1}E(s^2)$$

$$= E\left(\frac{n}{n-1}s^2\right)$$

$$= E\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right]$$

So the unbiased estimator of $\left\{\frac{1}{N}\sum_{j=1}^{N}(X_j - \bar{X})^2\right\}$ is $\left[\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right]$

Motivation

- Suppose we toss a coin with head probability p and we get a head.
- Between the following two options, which is more LIKELY ?
  1. p=0.7,
  2. p=0.9

Example

Number of toss	Number of head	option 1 for p	option 2 for p	Which one is more likely
100	90	0.2	0.8	option 2
10	1	0.2	0.5	option 1
100	100	0.6	0.9	option 2
20	4	0.8	0.3	option 2

Maximal Likelihood Estimator: Definitions

- Let $X_1, \ldots, X_n$ be i.i.d. with pdf $f_\theta(x)$.
- The *likelihood function* is defined by

$$L = L(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

- The *maximum likelihood estimator* or MLE, denoted by $\hat{\theta}$ is the value of θ that maximizes L.

Maximal Likelihood Estimator: Definitions (continued)

- Let $X_1, \ldots, X_n$ be i.i.d. with pmf $P_\theta(X)$.
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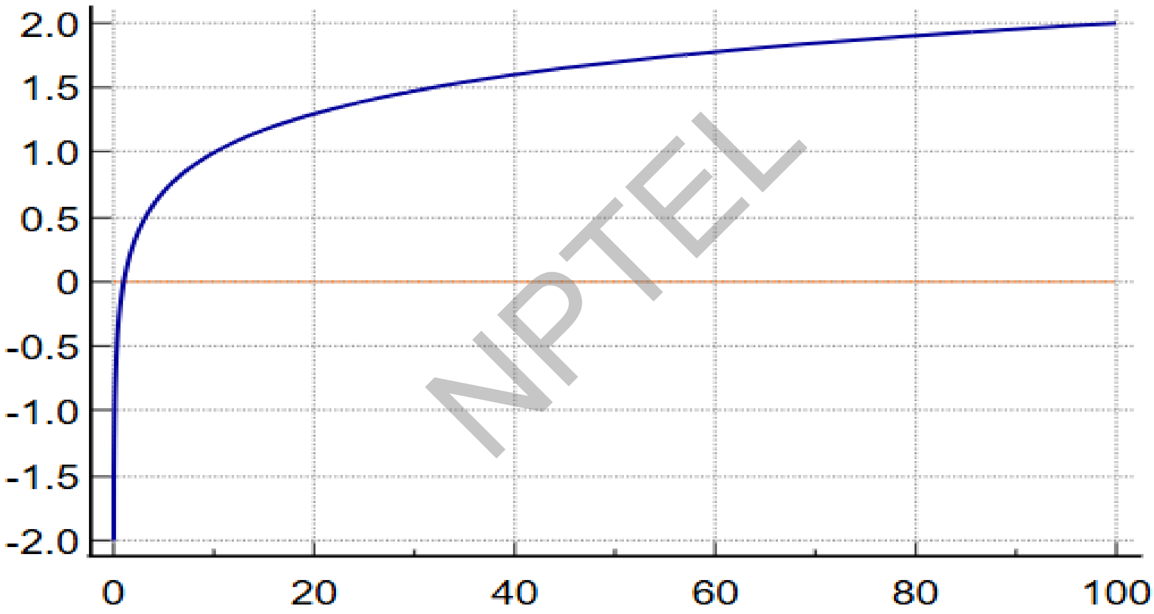
Graph of the logarithm function



Maximum Likelihood Estimation for the Exponential Distribution

- Recall that the pdf of the exponential distribution is given by

$$f_\theta(x) = \frac{1}{\theta} e^{\frac{-x}{\theta}}$$

  where $x \in [0, \infty)$ and $\theta > 0$
- Let $X_1, \ldots, X_n$ be random samples from exponential distribution.
- Our target is to estimate the parameter θ.

Maximum Likelihood Estimation for the Exponential Distribution (continued)

- The likelihood function is given by

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{n} f(X_i;\theta) \\ &= \frac{1}{\theta^n} e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i} \end{aligned}$$

- Taking log both sides we have

$$\begin{aligned} \log(L(\theta)) &= \log(\theta^{-n}) + \log(e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i}) \\ &= -n\log(\theta) - \frac{1}{\theta}\sum_{i=1}^{n} X_i \end{aligned}$$

Maximum Likelihood Estimation for the Exponential Distribution (continued)

- Now taking the partial derivative with respect to θ and set that equal to 0, we have

$$-\frac{n}{\theta} + \frac{\sum_{i=1}^{n} X_i}{\theta^2} = 0$$

$$\text{or, } \theta = \frac{\sum_{i=1}^{n} X_i}{n} = \bar{X}$$

- Therefore the maximum likelihood estimate of θ is $\hat{\theta} = \bar{X}$ and this gets verified by the second derivative test.

Maximum Likelihood Estimation for the Exponential Distribution (continued)

Also note that

$$\frac{\partial^2 \log(L)}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2\sum_{i=1}^{n} X_i}{\theta^3}$$
$$= \frac{n\theta - 2\sum_{i=1}^{n} X_i}{\theta^3}$$

Now when $\theta = \hat{\theta} = \bar{X}$, then the above expression yields,

$$\frac{\partial^2 \log(L(\theta))}{\partial \theta^2} = \frac{-n\bar{X}}{\theta^3} < 0$$

Therefore θ attains maximum at $\hat{\theta} = \bar{X}$

Examples: Continuous Distribution

Likelihood Estimation for Normal Distribution

- The pdf of the $\mathcal{N}(\mu, \sigma^2)$ distribution is given by

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

  for all $x \in \mathbb{R}$.
- Let $X_1, \ldots, X_n$ be a sample from $\mathcal{N}(\mu, \sigma^2)$, where both μ and σ^2 are unknown and $\theta = (\mu, \sigma^2)$.
- The likelihood function is

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^{n} f_\theta(X_i) \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} e^{\left\{-\sum_{i=1}^{n} \frac{(X_i-\mu)^2}{2\sigma^2}\right\}} \end{aligned}$$

Likelihood Estimation for Normal Distribution (continued)

- Taking log both sides we obtain

$$\log(L(\mu,\sigma^2)) = -\frac{n}{2}\log\sigma^2 - \frac{n}{2}\log(2\pi) - \left\{\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{2\sigma^2}\right\}$$
$$= -n\log(\sigma) - \frac{n}{2}\log(2\pi) - \left\{\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{2\sigma^2}\right\}.$$

- The likelihood equations are obtained by setting $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\mu} = 0$ and $\frac{\partial(\log L)}{\partial\sigma^2} = 0$

Likelihood Estimation for Normal Distribution (continued)

- $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\mu} = 0$ gives

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0$$

$$\text{or, } (X_1 + \cdots + X_n) - n\mu = 0$$

$$\text{or, } \mu = \frac{X_1 + \cdots + X_n}{n}$$

We get $\hat{\mu} = \bar{X}$

Likelihood Estimation for Normal Distribution (continued)

- To estimate σ^2, we need to set $\frac{\partial(\log(L(\mu,\sigma^2)))}{\partial\sigma^2} = 0$ i,e,

$$\frac{-n}{2}\frac{1}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i - \mu)^2 = 0$$

$$\text{or, } \frac{n}{2} = \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2$$

$$\text{or, } \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$$

$$= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

- Thus the likelihood estimations of μ and σ^2 are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$ respectively.
- One can check through the second derivative test that μ and σ^2 attain maximum at $\hat{\mu}$ and $\hat{\sigma}^2$

Maximum Likelihood Estimation for the Uniform $(0, \theta)$ Distribution

- Recall that the PDF is

$$f_\theta(x) = \begin{cases} \frac{1}{\theta}, & \text{if } 0 \le x \le \theta \\ 0, & \text{otherwise} \end{cases}$$

- Let $X_1, \ldots, X_n \sim \mathrm{Unif}(0, \theta)$.
- Consider a fixed value of θ. Suppose $\theta < X_i$ for some i. Then $f_\theta(x_i) = 0$ and hence the Likelihood function

$$L = \prod_{i=1}^{n} f_\theta(X_i) = 0.$$

Maximum Likelihood Estimation for the Uniform $(0, \theta)$ Distribution (continued)

- It follows that $L = 0$ if any $X_i > \theta$. Therefore $L = 0$ if $\theta < X_{(n)}$, where $X_{(n)} = \max\{X_1, \dots, X_n\}$.
- Now consider $X_{(n)} \leq \theta$. For every X_i we have that $f_\theta(X_i) = 1/\theta$.
- The likelihood estimator

$$L(\theta(x)) = \begin{cases} \frac{1}{\theta^n}, & \text{if } 0 \leq X_{(n)} \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

- Hence $\hat{\theta} = X_{(n)}$ maximizes it, so it is MLE.

Maximum Likelihood Estimation for the Gamma Distribution

- Recall that the pdf of the Gamma distribution can be written as

$$f_\theta(x) = \frac{1}{\Gamma(p)\theta^p} x^{p-1} e^{\frac{-x}{\theta}}$$

  where $x \in [0, \infty)$ and $p, \theta > 0$.
- Let $X_1, \ldots, X_n$ be random sample for the Gamma distribution.
- Our task is to derive the MLE of the parameter θ if p is known. where $x, p, \theta > 0$.

Maximum Likelihood Estimation for the Gamma Distribution

- In this case the likelihood function is

$$
\begin{aligned}
L(\theta) &= \prod_{i=1}^{n} f_\theta(X_i) \\
&= \prod_{i=1}^{n} \frac{1}{\Gamma(p)\theta^p} {X_i}^{p-1} e^{\frac{-X_i}{\theta}} \\
&= \theta^{-np}\Gamma(p)^{-n} e^{\frac{-1}{\theta}\sum_{i=1}^{n} X_i} \prod_{i=1}^{n} {X_i}^{p-1}
\end{aligned}
$$

Maximum Likelihood Estimation for the Gamma Distribution (continued)

- Taking log both sides we have,

$$\log(L(\theta)) = -np\log(\theta) - n\log(\Gamma(p)) - \frac{1}{\theta}\sum_{i=1}^{n} X_i + \log\prod_{i=1}^{n} {X_i}^{p-1}$$

- Now setting $\frac{\partial \log(L(\theta))}{\partial\theta} = 0$ we get

$$\frac{-np}{\theta} + \frac{\sum_{i=1}^{n} X_i}{\theta^2} = 0$$
$$\text{or, } \frac{\sum_{i=1}^{n} X_i}{np} = \theta$$

i.e, $\hat{\theta} = \frac{\bar{X}}{p}$ is the MLE of θ.

- One can check through second derivative test that θ attains maximum at $\hat{\theta}$.

Likelihood Estimation for Discrete Distribution

Likelihood Estimation for the Poisson μ population

- The pdf of the Poisson μ population

$$P_\mu(X = x) = e^{-\mu}\frac{\mu^x}{x!}$$

where $x \in \mathbb{N}_0$.

- Let $X_1, \ldots, X_n$ be sample from Poisson μ population.
- The likelihood function is

$$\begin{aligned} L(\mu) &= \prod_{i=1}^{n} P_\mu(X_i) \\ &= e^{-n\mu}\frac{\mu^{n\bar{X}}}{X_1! \ldots X_n!} \end{aligned}$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Likelihood Estimation for the Poisson μ population (continued)

- Taking log both sides,

$$\log(L(\mu)) = -n\mu + n\bar{X}\log(\mu) - \log(X_1! \dots X_n!).$$

- Now taking the partial derivative with respect to μ and setting that equal to zero we get

$$-n + \frac{n\bar{X}}{\mu} = 0$$
$$\text{or, } \bar{X} = \mu$$

- Thus the maximum likelihood estimate for μ is $\hat{\mu} = \bar{X}$.
- One can check through second derivative test that μ attains maximum at $\hat{\mu}$.

Likelihood Estimation for the Bernoulli Distribution

- Recall that the probability mass function of the Bernoulli (p) is

$$P_p(X = x) = p^x(1-p)^{1-x}$$

for $x = 0, 1$ where p is the probability of success in each trial and the unknown parameter is p.

- Let $X_1, \ldots, X_n \sim$ Bernoulli (p) where p is the probability of success in each trial. The likelihood function is

$$L = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^S(1-p)^{n-S}$$

where $S = \sum_{i=1}^{n} X_i$.

Likelihood Estimation for the Bernoulli Distribution (continued)

- After taking log both sides,

$$\log(L) = S\log(p) + (n - S)\log(1 - p)$$

- Now take the derivative and set it equal to 0, we have

$$\begin{aligned} \frac{S}{p} - \frac{n-S}{1-p} &= 0 \\ \text{or, } (1-p)S - (n-S)p &= 0 \\ \text{or, } S - pS - np + pS &= 0 \\ \text{or, } S &= np \end{aligned}$$

- Therefore, $\hat{p} = \frac{S}{n} = \bar{X}$ which is the required MLE of p.
- One can check via second derivative test that p attains maximum at $\hat{p}$.

Likelihood Estimation for the Binomial (N, p) Distribution

- Recall that the pdf of the binomial (n, p) population is

$$P_p(X = x) = \binom{N}{x} p^x (1-p)^{N-x}$$

  N is the number of trials and p is the probability of success in each trial and x is the number of total successes.
- Let $X_1, \ldots, X_n \sim \mathrm{Bin}(N, p)$.
- Here our task is to estimate the parameter p.

Likelihood Estimation for the Binomial (N, p) Distribution (continued)

- Therefore the likelihood function is

$$
\begin{aligned}
L(p) &= \prod_{i=1}^{n} P_p(x_i) \\
&= \binom{N}{X_1} \dots \binom{N}{X_n} p^{X_1+\dots+X_n}(1-p)^{nN-(X_1+\dots+X_n)}
\end{aligned}
$$

- Taking log both sides

$$
\log(L(p)) = (X_1+\dots+X_n)\log(p) + \{nN - (X_1+\dots+X_n)\}\log(1-p) + \log\left(\binom{N}{X_1}\dots\binom{N}{X_n}\right)
$$

Likelihood Estimation for the Binomial (N, p) Distribution (continued)

- Now $\frac{\partial \log(L(p))}{\partial p} = 0$ gives

$$\begin{aligned}
\frac{X_1 + \cdots + X_n}{p} &= \frac{nN - (X_1 + \cdots + X_n)}{1 - p} \\
&= \frac{nN - (X_1 + \cdots + X_n) + (X_1 + \cdots + X_n)}{p + 1 - p} \\
&= nN \\
\text{or, } p &= \frac{(X_1 + \cdots + X_n)}{nN} \\
&= \frac{\bar{X}}{N}
\end{aligned}$$

i,e, $\hat{p} = \frac{\bar{X}}{N}$ is the required maximum likelihood estimate of p.

- One can check through second derivative test that p attains maximum at $\hat{p}$.

Example

Question: Let us choose a sample X_1 of unit size from a population having pdf:

$$f_\alpha(x) = \begin{cases} \frac{2}{\alpha^2}(\alpha - x) & \text{if } 0 < x < \alpha \\ 0 & \text{otherwise} \end{cases}$$

Prove that the maximum likelihood estimate of the parameter α is $2X_1$.

Example (continued)

Solution: For a sample of unit size, the likelihood function is

$$\begin{aligned} L(\alpha) &= f_\alpha(X_1) \\ &= \frac{2}{\alpha^2}(\alpha - X_1) \end{aligned}$$

Taking log both sides of the above equation we get

$$\log(L(\alpha)) = \log(2) - 2\log(\alpha) + \log(\alpha - X_1)$$

Example (continued)

Now $\frac{\partial \log(L(\alpha))}{\partial \alpha} = 0$ gives

$$\begin{aligned} \frac{-2}{\alpha} + \frac{1}{\alpha - X_1} &= 0 \\ \text{or, } 2(\alpha - X_1) - \alpha &= 0 \\ \text{or, } \alpha &= 2X_1 \end{aligned}$$

Also

$$\frac{\partial^2 \log(L(\alpha))}{\partial^2 \alpha} = \frac{2}{\alpha^2} - \frac{1}{(\alpha - x)^2}$$

At $\alpha = X_1$, its numerator becomes

$$2(2X_1 - X_1)^2 - (2X_1)^2 = -2{X_1}^2 < 0$$

Hence the MLE of α is $\hat{\alpha} = 2X_1$.

References

1. https://egrcc.github.io/docs/math/all-of-statistics.pdf
2. An Introduction To Probability And Statistics, 3rd edition by V.K. Rohatgi and A.K. Saleh, Wiley Series in Probability and Statistics.
3. Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.


Thank you

SWAYAM NPTEL COURSE ON

STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

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Module 02: Statistical Inference 1

Lecture 04: Method of Moments

Philosophy

Sample moments resemble population moments.

Solved Example

Example 1

Question: A random variable X takes the values with respective probabilities as below :

$$P(X=0) = \frac{\theta}{4N} + \frac{1}{2}(1 - \frac{\theta}{N})$$
$$P(X=1) = \frac{\theta}{2N} + \frac{\alpha}{2}(1 - \frac{\theta}{N})$$
$$P(X=2) = \frac{\theta}{4N} + \frac{1-\alpha}{2}(1 - \frac{\theta}{N})$$

where N is a known number and α, θ are unknown parameters. If 75 independent observations on yielded the values $0, 1, 2$ with frequencies 27,38,10 respectively. Estimate θ.

Example 1 (continued)

$$
\begin{aligned}
E(X) &= 0 \cdot \left[\frac{\theta}{4N} + \frac{1}{2}\left(1 - \frac{\theta}{N}\right)\right] + 1 \cdot \left[\frac{\theta}{2N} + \frac{\alpha}{2}\left(1 - \frac{\theta}{N}\right)\right] + 2\left[\frac{\theta}{4N} + \frac{1-\alpha}{2}\left(1 - \frac{\theta}{N}\right)\right] \\
&= \frac{\theta}{N} + \left(1 - \frac{\theta}{N}\right)\left[\frac{\alpha}{2} + (1-\alpha)\right] \\
&= \frac{\theta}{N} + \left(1 - \frac{\theta}{N}\right)\left(1 - \frac{\alpha}{2}\right) \\
&= 1 - \frac{\alpha}{2}\left(1 - \frac{\theta}{N}\right)
\end{aligned}
$$

and

$$
\begin{aligned}
E(X^2) &= 0^2 \cdot \left[\frac{\theta}{4N} + \frac{1}{2}\left(1 - \frac{\theta}{N}\right)\right] + 1^2 \cdot \left[\frac{\theta}{2N} + \frac{\alpha}{2}\left(1 - \frac{\theta}{N}\right)\right] + 2^2\left[\frac{\theta}{4N} + \frac{1-\alpha}{2}\left(1 - \frac{\theta}{N}\right)\right] \\
&= \frac{3\theta}{2N} + \left(1 - \frac{\theta}{N}\right)\left(2 - \frac{3\alpha}{2}\right) \\
&= 2 - \frac{\theta}{2N} - \frac{3}{2}\alpha\left(1 - \frac{\theta}{N}\right)
\end{aligned}
$$

Example 1 (continued)

We estimate first population moment with first sample moment

$$1 - \frac{\alpha}{2}\left(1 - \frac{\theta}{N}\right) = \frac{1}{75}(1 \cdot 38 + 2 \cdot 10) = \frac{58}{75}$$

We also estimate second population moment with second sample moment

$$2 - \frac{\theta}{2N} - \frac{3}{2}\alpha\left(1 - \frac{\theta}{N}\right) = \frac{1}{75}(1^2 \cdot 38 + 2^2 \cdot 10) = \frac{78}{75}.$$

Example 1 (continued)

We need to solve the following system of equations

$$1-\frac{\alpha}{2}\left(1-\frac{\theta}{N}\right)=\frac{58}{75}$$

$$\text{and, } 2-\frac{\theta}{2N}-\frac{3}{2}\alpha\left(1-\frac{\theta}{N}\right)=\frac{78}{75}$$

The first equation gives

$$\frac{\alpha}{2}(1-\frac{\theta}{N})=1-\frac{58}{75}=\frac{17}{75}.$$

Example 1(continued)

Putting the above value in the second equation we obtain

$$\begin{aligned} 2 - \frac{\theta}{2N} - 3 \times \frac{17}{75} &= \frac{78}{75} \\ \text{or, } \theta &= \frac{42}{75} N \end{aligned}$$

Finally

$$\begin{aligned} \frac{\alpha}{2}\left(1 - \frac{42}{75}\right) &= \frac{17}{75} \\ \text{or, } \alpha &= \frac{34}{33} \end{aligned}$$

Therefore $\hat{\theta} \sim \frac{42}{75} N$ and $\hat{\alpha} \sim \frac{34}{33}$.

Example 2

Question: Let $X_1 = 1, X = 2, X_3 = 7$ and $X_4 = 10$ are four random samples picked from a normal distribution of mean μ and variance σ^2. Estimate μ and σ^2.

Example 2 (continued)

Solution: In our sample

$$\begin{aligned} E(X) &= \bar{X} \\ &= \frac{1+2+7+10}{4} \\ &= 5 \end{aligned}$$

Also, the second-order sample moment

$$\begin{aligned} E(X^2) &= \frac{1^2+2^2+7^2+10^2}{4} \\ &= \frac{154}{4} \\ &= 38.5 \end{aligned}$$

Example 2 (continued)

We continue with the same philosophy as before. We know population variance

$$\sigma^2 = E(X^2) - \{E(X)\}^2$$
$$= E(X^2) - 25$$

We have the pair of equation

$$\mu = 5$$
$$\text{and, } \sigma^2 = 38.5 - 25 = 13.5$$

Therefore the estimates of μ and σ^2 are $\hat{\mu} \sim 5$ and $\hat{\sigma}^2 \sim 13.5$ respectively.

Methods of Moments

Suppose that the parameter $\theta = (\theta_1, \theta_2, \ldots, \theta_k)$ has k components. For $1 \leq j \leq k$, we define the j-th moment as

$$\alpha_j = \alpha_j(\theta) = E_\theta(X^j)$$

and the j-th sample moment

$$\hat{\alpha}_j = \frac{1}{n} \sum_{i=1}^{n} X_i^j$$

Definiton

The *method of moments estimator* $\hat{\theta}$ is defined to be the value of θ such that

$$\begin{aligned}
\alpha_1(\hat{\theta}) &= \hat{\alpha_1} \\
\alpha_2(\hat{\theta}) &= \hat{\alpha_2} \\
&\vdots \\
\alpha_k(\hat{\theta}) &= \hat{\alpha_k}
\end{aligned}$$

Moments Estimator for Continuous Distributions

Normal (μ, σ^2) Distribution

- Let $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$.
- Parameters are given by the tuple $\theta = (\mu, \sigma^2)$
- Then $\alpha_1 = E_\theta(X_1) = \mu$ and $\alpha_2 = E_\theta({X_1}^2) = \sigma^2 + \mu^2$

  [Recall that, $Var(X) = E(X^2) - \{E(X)\}^2$ and hence $E(X^2) = Var(X) + \{E(X)\}^2$].

Normal (μ, σ^2) Distribution (continued)

- Now we need to solve the equations

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

and

$$\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum_{i=1}^{n} {X_i}^2.$$

Normal (μ, σ^2) Distribution (continued)

- From the first equation we get

$$\hat{\mu} = \bar{X}.$$

- From the second equation we get

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \hat{\mu}^2$$

Normal (μ, σ^2) Distribution (continued)

- From the first equation, substituting, $\hat{\mu} = \bar{X}$ we have

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2$$
$$= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

- Therefore, by the method of moments, the estimates of μ and σ are $\bar{X}$ and $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$ respectively.

Exponential Distribution

- Let $X_1, \ldots, X_n$ be the samples from the exponential distribution whose pdf is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0. \\ 0 & \text{otherwise} \end{cases}$$

- The parameter is λ what we need to estimate.
- Remember that $\alpha_1 = E(X_i) = \frac{1}{\lambda}$ for $1 \leq i \leq n$ and $\hat{\alpha_1} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$.
- Equating them, we get

$$\frac{1}{\lambda} = \bar{X}$$
$$\text{or, } \hat{\lambda} = \frac{1}{\bar{X}}$$

- Thus the estimate of λ is $\frac{1}{\bar{X}}$.

Moments Estimator for Discrete Distributions

Bernoulli (p) Distribution

- Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$.
- Here p, the probability of success at each trial, is the parameter.
- Then $\alpha_1 = E_p(X) = p$ and $\hat{\alpha}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$.
- By equating these, we get the estimator

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

References

References

1. https://egrcc.github.io/docs/math/all-of-statistics.pdf
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